

The State-of-the-Art Techniques of Artificial Intelligence for Smart Grids

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Abstract— The Smart Grid represents a new paradigm for electricity generation and distribution. Shifting from the traditional power grid to this innovative system will provide better management of energy demands and address other issues in the power sector. We explore various domains of this paradigm to provide insight into the topic, discussing major issues that arise during the transition. The application of Artificial Intelligence is already proving its impact on different challenges within the Smart Grid and can be further implemented for other existing problems. We discuss various methods to improve its performance, presenting AI techniques that can handle numerous issues. Future directions for each concern are also presented.

Keywords—Smart Grid, Power Grid, Energy, Artificial Intelligence, Reinforcement learning.

I. INTRODUCTION

The term grid is generally used for electricity system that broadly supports four operations i.e. generation, transmission, distributions and control. Power grid is an interconnected network for transmission of electricity, and further its distribution to the consumers. The target of electricity generation is to cater the demand of the energy. The energy needs to be generated only up to the demand. In case more energy is generated, it need to be stored in the form of batteries which is a costly affair. The energy generation process of now days is shifting from fossil fuels towards renewable energy such as wind and solar [1]. The renewable energy generation is also possible at small sites rather than only at huge energy plants like coal plants. This shift requires new methods for managing the energy flow, as now it is not only produced by the energy companies but also by standalone and small energy producers. As different sources are generating different capacity of electricity, they need to be manage with better grids in future.

The term Smart Grid (SG) is used to represent a network of energy distribution, where digital communication technology is used in a Two-way flow of data and energy [2]. Basically, SG is an electricity network. It is a developing network that contains transmission lines, equipment's, control and new technology working collaboratively to cater the need of this century. The paradigm of SG allows utility companies to optimize their generation, transmission, distribution and management of electricity. The pro-consumers (i.e. consumers that also produce energy) get more detail understanding of the energy they used and produce. Two-way flow of information and electricity is in the foundation of SG. In fact, Two-way exchange of information between utility grid and consumers is the basic difference of power grid and SG.

SG are equipped with sensors, controllers, switches and substations that are capable of strong communication. It is also capable of monitoring the activities of the grid connected system and provides the real time information of all events occurring in the power system. The major components of SG system are Smart power meter, Smart substations, Smart distribution, Smart generation, Intelligent appliances, different sensors and automation. In SG, the electric power can be generated and distributed from multiple generation plants. The advance metering system of SG even utilize the fiber optic routers for advance metering infrastructure. The circuit breakers and distribution routers are also futuristic. SG can detect respond and react to various issues, giving it self-healing capabilities.

The consumers now days' act as produces at the same time. This new generated energy can either be stored at consumer's place or need to be forwarded to distribution center of power companies for further forwarding at the required location [3]. Installing the similar infrastructure that was installed in early era of power grid is not suitable, as this time we need to further transfer some information along with energy generated to make some decision capabilities at the distribution centers. Moreover, the integration of renewable energy is possible using SG. We need to generate energy with the combination of different technologies. Generation of energy cannot only rely only on renewable energy, as they are in limited scope, also they depend on weather conditions rather than demand conditions. Somewhere, the shift towards this new paradigm is beneficial to our society, as it reduces the impact on environment such as emission of CO₂ and global warming. SG is going to promote renewable energy system. Dependency on larger grid and transportation losses will also reduce in this new paradigm. SG create a better demand supply chain system. It reduces the wastage and storage of energy, as traditional grid does.

The first step toward this shift is smart digital meter that have capability to import and export energy, as compared to traditional energy meter that can only imports energy. Smart meters also have various sensors, that send real time data to utility. Sensors are even installed in distribution transformation and substation, and finally to generating station. The existing structure of power grid is centralized but SG is considered as decentralized. Figure 1 depicts the major domains based upon which the traditional power grid is differentiated to SG.

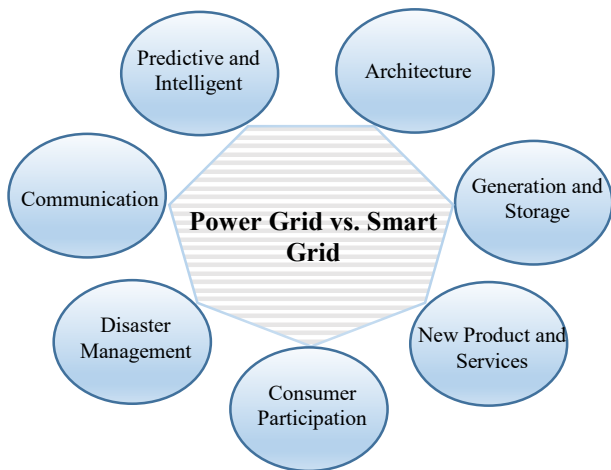


Figure1: Domains where SG varies from Traditional grid

The remainder of this paper is organized as follows. Section 2 presents an overview of Artificial Intelligence (AI) and its various techniques. Section 3 discuss existing challenges of SG and their solution from AI. In section 4, we discuss future of SG with AI and finally, section 5 conclude the article.

II. AI FOR SMART GRID

In its most basic form, AI comprises a collection of techniques designed to imbue networks with intelligence akin to human capabilities. Its capabilities extend beyond the realm of computer science to encompass other fields as well. It is reshaping the way interactions occur in the world. Furthermore, it encompasses specific areas such as Machine Learning (ML), Neural Networks (NN), Computer Vision, Deep Learning, Robotics, Fuzzy Logic, Natural Language Processing, and Expert Systems. AI enables systems to learn without the need for rewriting programs. They can predict and make decisions once programmed [4].

The power grid is the driving force that lights our homes, recharges our electric vehicles (EVs), and powers industries, among other things. AI is impacting various fields, spanning from healthcare to finance, and now it is making waves in the energy sector. The substantial volume of data produced by smart grids, combined with the complexities of energy distribution, renders this field ideal for the application of AI. Techniques of AI like ML, Internet of Things (IoT), in combination with cloud connections, are providing advancements to SG. With the help of such advanced techniques, the grids are capable of managing more power generation and its efficient distribution. There is an imbalance in the peak production and consumption due to the unavailability of renewable resources all the time. AI can serve as a powerful tool to address such challenges of SG [5]. It can optimize energy flow, minimize or eliminate wastage in distribution, balance supply and demand, enhance precision, improve operational efficiency, and lay the groundwork for renewable energy infrastructure. It can also automate few processes and therefore require less human intervention, ultimately increasing efficiency. This also reduces the chances of human error; therefore, we can say that it is a reliable energy distribution. If connected with a weather forecasting system, AI can predict demand forecasting, patterns can also be considered to predict the demand. In essence, AI tools are necessary to optimize profits

for both consumers and producers. The use of AI in the SG paradigm will address its challenges by replacing inequities between production and consumption load, which results in the optimization of the SG network. The data related to energy consumption, weather patterns, and grid state are very useful for AI. It makes this data useful to producers and consumers by applying its algorithms. The implementation of AI with SG represents a promising route towards a more ecological, energy-efficient, and low-carbon future solution. Various data analytic algorithms can be developed to ensure that batteries charge and discharge exactly at specific times based on various network constraints and pricing equations. The current state of AI and smart grids is riddled with obstacles. Inefficiencies, limitations, and demands of SG compel further innovation and research from AI.

III. ISSUES IN SMART GRID

A. Load Forecasting (Issue 1)

Load Forecasting (LF) involves the estimation or prediction of future electricity consumption by consumers. It plays a pivotal role in energy systems, facilitating efficient management of energy generation, transmission, and distribution. Several factors must be considered before making forecasts, including weather conditions, special events, time of day, seasonal variations, and day of the week. With the integration of diverse renewable energy generation methods, LF has become increasingly crucial. Moreover, it significantly impacts various aspects of SG operations, including reliability, stability, resource allocation, and demand management. Consequently, LF is recognized as a paramount concern within the realm of SG.

Various methods for LF includes:

- *Time series analysis*: This technique uses historical load data and suitable for Short Term Load Forecasting (STLF) i.e. LF for few hours to few days.
- *Advance Neural networks*: This method provide more accurate forecasting capable of adapt as per changing load patter. It also incorporates various other factors like weather conditions.
- *Support vector machine*: It is one of the most effective technique for LF that can work with limited amount of available training data. It captures nonlinear relationships and complex patterns and therefore do accurate forecast.
- *Machine learning methods*: Ensemble methods combines various available techniques to increase the prediction accuracy. ML techniques are combined to provide a reliable forecast.
- *Deep learning models*: These models are capable of capturing long term dependencies and also learn automatically hierarchal representation of input features. thus, reduces the need for manual input. It is very useful for Mid Term Load Forecasting (MTLF) i.e. forecasting that predicts for a day to year).
- *Hybrid models*: These models integrate the benefits of various techniques, creating a hybrid model that amalgamates different approaches. Such models offer enhanced robustness and accuracy in predictions compared to single-area models.

Each of the methods described above has its own unique set of benefits and constraints. AI-driven approaches to LF provide advantages in handling intricate relationships, adaptability, scalability, enhanced precision, real-time

processing, and dynamic forecasting. Research conducted utilizing a deep learning model employing stacked denoising auto-encoders, coupled with the training of a support vector regression model, has been documented in [6]. This approach yields more precise electricity load forecasts with reduced errors. Furthermore, a feature extraction model is employed to enhance the refinement of abstract data from heterogeneous sources. [7] introduces an ensemble incremental learning technique designed for managing STLF. This method combines Discrete Wavelength Transform, Empirical Mode Decomposition, and random vector linking in a hybrid approach. Its efficacy is showcased through comparative analysis with diverse benchmark methods. In [8], a fusion of three distinct algorithms—Improved Ensemble Empirical Mode Decomposition, Grey Wolf Optimization (GWO), and Multiple Kernel Extreme Learning (MKEL)—is employed for STLF. MKEL is utilized for forecasting, while GWO optimizes the process. Evaluation outcomes demonstrate superior performance across multiple assessment criteria in comparison to alternative algorithms. In [9], a hybrid load forecasting model is proposed, integrating Kalman Filtering, Wavelength, and Artificial Neural Network (ANN) methods to enhance efficiency. Tailored for STLF, it boosts overall system performance and quality. Simulation results across diverse datasets validate its depicted performance. In [10], a technique for MTLF is introduced, which integrates a novel approach combining Multilayer Perceptron NN with optimization techniques. Specifically, the algorithm employs Particle Swarm Optimization (PSO) and an enhanced version of the Ant Lion Optimizer to enhance forecasting accuracy. In [11], a model is introduced that utilizes a "Quantum Controlled-Not Gate Induced Feed-forward Neural Network" and optimizes it with the Quantum Adaptive Differential Evolutionary learning algorithm for load forecasting. Additionally, the article addresses demand-supply management needs and demonstrates improvements in root-mean-square error. In [12], the article presents a model tailored for STLF within smart grids, particularly emphasizing solar-based grids. It employs a probabilistic approach for uncertainty quantification, training the Lower Upper-bound Estimation Method with real-time data from smart grids. ML algorithms are employed to analyse data from smart grids, and the White Shark Optimization algorithm is utilized for training within a heuristic environment. In [13], an efficient probabilistic STLF method based on deep learning is introduced. This method employs a Multilayer Perceptron (MLP)-mixer structure for forecasting, coupled with random quantile regression trained on a random variable. Through the fusion of these two algorithms, a novel model called Quantile-mixer is proposed. Comparative analysis against other deep learning-based algorithms highlights the effectiveness of this approach in load forecasting. Deep neural networks and deep learning techniques are gaining prominence for LF in SG, surpassing other methods. In [14], a hybrid algorithm integrating Beluga Whale Optimization (BWO) with Deep Learning is introduced. This hybrid approach is tailored for STLF in SG. It incorporates a zero-score normalization technique for pre-processing input datasets and employs Convolutional Bidirectional Long Short-term Memory (LSTM) alongside auto-encoders for predictions. BWO is utilized for the optimal selection of the convolution bidirectional algorithm, enhancing overall prediction accuracy.

B. Fault Protection (Issue 2)

In smart grids, a fault refers to any deviation or disruption from the normal operating conditions of the electrical system. This disruption can affect the flow of electricity and may impact the performance of various components within the grid. Faults can occur due to various reasons such as equipment failure, environmental factors, cyber-attacks, or human errors. Some common types of faults in SG systems include: Open Circuit, Short Circuit, Ground Fault, Phase to Phase Fault, Phase to Ground, Intermittent fault, and Transient Fault. Specific to smart grids, there are additional faults such as Cyber-attacks, stemming from unauthorized access, disrupting grid operations, and Communication Faults, originating from communication infrastructure issues, which limit monitoring capabilities. These faults can result in equipment damage, protective device tripping, power outages, safety hazards, and synchronization issues among interconnected grids. Effectively addressing faults in smart grids necessitates diverse monitoring, detection, and mitigation strategies. Real-time monitoring, predictive analysis, and automated control systems are essential for handling faults to uphold grid reliability, resilience, and security.

A methodology is devised to detect evolving and cross-country high impedance faults utilizing Wavelet Packet Transform (WPT) and Extreme Learning Machine (ELM) techniques [15]. The WPT is employed to isolate the high-frequency coefficient of the three-phase current, with an AI-driven classifier then utilized to identify the phase affected by the fault. Comparative analysis suggest that this solution offers resilient and dependable fault detection capabilities. In [16], a proposal introduces a ML approach combining discrete wavelet transform (DWT) with double-channel extreme learning. This method aims to pinpoint and categorize faults occurring in transmission lines. This proposal demonstrates faster fault detection times coupled with enhanced accuracy. In [17], switch values, including relay protection actions, serve as diagnostic indicators for faults in the power system. A stacked sparse auto-encoder is leveraged to diagnose line trip faults based on analog operational data. This method stands out as a highly effective approach to data learning, owing to its extensive utilization in training mechanisms. Combining stacked sparse auto-encoder with support vector machine and principal component analyses ensures accurate fault diagnosis within the grid system. In [18], a ML based technique utilizing Support Vector Machine (SVM) is implemented for fault detection within the grid. The SVM-based algorithm excels in precision, particularly in islanding detection, which occurs when a section of power becomes isolated from the rest of the grid. The evaluation of non-detection zones and detection time is conducted using the active energy generated and distribution network voltage levels. Comparative analysis against existing methods demonstrates superior accuracy and precision in islanding and grid fault detection. [19] introduces the deployment of an AI core within Alveo Acceleration, a high-performance computing solution developed by a leading provider of programmable logic devices. The proposal focuses on training an AI model for rapid fault detection in Smart Grids, with implementation conducted on Field Programmable Gate Arrays (FPGAs). Comparative analysis with Graphics Processing Units (GPUs) reveals that the proposal consumes less energy and offers reduced latency. Automating fault detection within Smart Grids is imperative. Article [20]

introduces an Ensemble Deep Learning technique aimed at automation, validation, and addressing overfitting concerns. Firstly, data is pre-processed through operations on raw data, followed by the execution of PSO. Subsequently, an ensemble system is deployed, incorporating various deep learning techniques. The final decision is determined by a majority voting engine within the ensemble system. In [21], a multi-layered fault protection scheme is presented for medium voltage distribution systems. This algorithm utilizes AI to communicate with Protective Devices (PDs) within the grid. Additionally, another algorithm facilitates communication among PDs on the same line to offer backup protection. Furthermore, an AI-based algorithm utilizing local data from each PD is proposed for fault protection and safeguarding against cyber-attacks. This algorithm assesses whether a fault exists within the system. Upon fault detection, protective measures in accordance with the grid's policy are promptly activated. Article [22] introduces the Transmission Line Fault location and classification (TLFed), a federated learning-based algorithm designed for fault location and classification. TLFed employs a 1-D convolutional NN and long-short term memory to enhance performance. By decentralizing the system, the proposal aims to bolster security and improve the effectiveness of fault diagnosis. TLFed exhibits high performance through its capability to accurately localize and detect faults within a SG.

C. Security (Issue 3)

As the paradigm and technological landscape shift towards Smart Devices (SD), the SG becomes increasingly susceptible to a multitude of attacks. The need for heightened security measures within the SG, unlike conventional power grids, arises from the significant difference in potential threats. While physical threats are conceivable in traditional power grids, the advent of SG introduces the ominous possibility of cyberattacks executed from virtually any corner of the globe. Cyberattacks targeting the SG can yield devastating consequences, including operational failures, synchronization losses, power supply disruptions, denial of service, data breaches, complete blackouts, and hardware damage. These attacks may involve injecting falsified data to deceive the system or inundating it with excessive data to overwhelm its capacities. Such methods represent common avenues for cyber assailants aiming to compromise the integrity and functionality of the SG. Effectively managing the security challenges within the SG demands a comprehensive, multi-layered strategy. Approach encompasses encryption, authentication methods, intrusion detection systems, security protocol etc. Figure 2 depicts major security concerns.

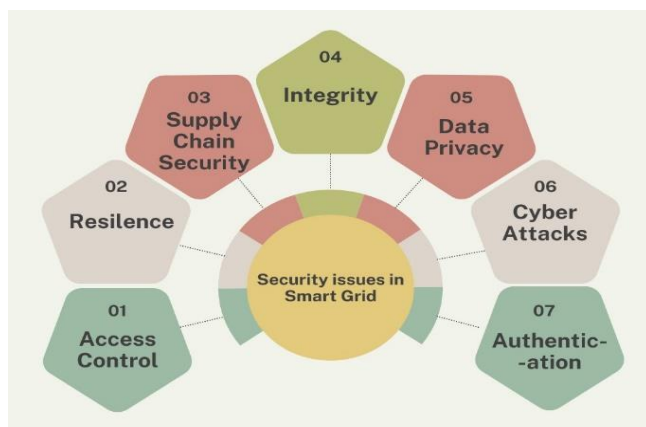


Figure 2: 7 Major security concerns in Smart Grid

[23] explores cyberattacks targeting Smart Meters (SM), an essential component in implementing the security grid. The proposed method relies on dimensional augmentation using recurrence plots and Visual Geometer Group Network (VGGNet). Real-time one-dimensional time series data from SM is collected and augmented into two-dimensional image data. Recurrence plots are employed to create this image data, which offers visually distinguishable features. These features serve as inputs to a Computer Vision-based algorithm, leveraging VGGNet techniques to discern data patterns and detect any potential contamination within the meter. [24] highlights the vulnerability of labelled datasets, crucial for training and testing within the SG, to malicious attacks that can go undetected. To address this, a resilient AI-based detector is proposed, specifically designed to identify such dataset attacks. This method focuses on capturing both spatial topological aspects and temporal correlations inherent in the measurement data. A neural network-based anomaly detection approach is introduced, demonstrating effectiveness against False Data Injection Attacks (FDIA) and data poisoning attempts. This approach relies solely on a single dataset for training and utilizes auto-encoders alongside a Chebyshev graph convolution recurrent layer with an attention mechanism, enabling the capture of spatial and temporal correlations within the measurement data. [25] focuses on enhancing security in hardware communication and data transmission for smart energy meters. It introduces the integration of LoRaWAN technology into the SG network design. Various sensors are deployed and by incorporating LoRaWAN technology, the design aims to prevent attacks on the Utility Centre (UC) by isolating the customer network from the UC. Data collected from the SM are encrypted and transmitted via LoRaWAN connections to data concentrators, which include essential information such as timestamps, SM identifications, power consumption data, GPS coordinates, humidity, and temperature readings. Furthermore, the redundant data concentrators facilitate the transmission of data to the utility network, enabling the establishment of an accurate control system for the SG. In [26], an intrusion detection model is introduced, capable of classifying binary, triple, and multi-class cyber-attacks. The model employs a Whale Optimization Algorithm (WOA) to initialize and adjust the weight factors of an Artificial Neural Network (ANN) to minimize mean square error. This model effectively addresses issues related to failure prediction, attacks, and failure detection. Each search agent within WOA optimizes a candidate network. The model pre-processes received data through cleansing, normalization, and feature selection. Data is divided using Gaussian random distribution for testing. Feature selection is integrated into the WOA-ANN classification process, and the ANN classification model, equipped with optimal weights and biases from training, is evaluated with testing data to assess model accuracy. The article [27] addresses electricity theft in the SG by analysing data to detect and classify power theft. It introduces an active learning-based ML model, utilizing various ML methods including Active Learning-based Random Forest, Extreme Gradient Boosting, K Nearest Neighbor, Category Boosting, and Light Gradient Boosting Machine classifiers. Classifiers implement active learning methodology by creating active learning objects, and their average scores are evaluated on test data. This approach is beneficial in scenarios with limited labelled data, requiring iterative labelling to improve classification performance. [28] introduces a scalable network approach for counter attacks, combining neural networks with

large combinatorial action approximations. Game theoretic methods transfer knowledge from small to large graphs. Deep Reinforcement Learning (RL) is utilized for combinatorial optimization over graphs, with a neural network model in a 2-player game setting. Tabular RL methods handle small, discrete state and action spaces, while a neural network model predicts utility for attacking or defending nodes in a graph using node embedding and minimax Q-learning. The RCR sampling method approximates large action combinations. This enables learning of attack and defence strategies on large-scale networks. [29] offers a framework for assessing the impact of False Data Injection Attacks (FDIA) on smart meters within the SG. It establishes propagation and attack models for FDIA, particularly targeting advanced metering infrastructure. The paper proposes a defence mechanism considering probabilistic code propagation, utilizing optimal code theory to devise optimal repair and path schemes for smart meters. A novel Federated Learning (FL) approach for SG anomaly detection is proposed [30], allowing local training of ML models on smart meters without sharing data with a central server to preserve privacy and security. SSL/TLS protocols safeguard the FL process against cyberattacks, ensuring data confidentiality, integrity, and authentication. The approach's effectiveness is demonstrated through comparisons with centralized models using various datasets and performance metrics such as precision, recall, and accuracy scores.

D. Energy Management (Issue 4)

Energy management in smart grids involves optimizing electricity generation, transmission, distribution, and consumption through advanced technologies like smart meters, demand response programs, integration of renewable energy sources, real-time monitoring, predictive analytics, grid optimization, cybersecurity, and customer engagement. The goal is to enhance efficiency, reliability, and sustainability while empowering consumers to make informed decisions about their energy usage. AI in smart grids optimizes various operations and engages customers effectively, leading to efficient, reliable, and sustainable grid operation.

In Article [31], a distributed framework is proposed for managing power systems. The framework is demonstrated using a simulated Nano grid to showcase its utility. Various power control strategies are introduced within the framework to mitigate the impact of power generation variability and fluctuating consumer loads. The proposed framework is evaluated using algorithms such as Ant Colony Optimization, PSO, and Genetic Algorithms. The findings suggest that the framework, particularly when utilizing PSO, is effective compared to alternative implementations. Additionally, fluctuating energy scenarios are simulated across Nano grids of varying sizes, ranging from small to large numbers of devices. In [32], an energy management scheme focusing on the integration of photovoltaics and storage systems is addressed. The study considers a multi-storey building with a high density of residences as a case study. The key components include a photovoltaic model and a battery energy storage system. A ML-based algorithm is proposed to forecast future energy demands. A software system monitors the storage system's status and receives live data regarding energy demand. Real-time consumption data is utilized by an AI module to predict future demand. Finally, all collected information is processed by an optimizer module, which suggests strategies for energy management.

Article [33] introduces a novel framework aimed at managing energy and reducing electricity bills, while also lowering the peak-to-average ratio. The framework utilizes an artificial neural network (ANN) based forecast engine and a modified enhanced differential evolution algorithm inspired by the GWO. The forecast engine predicts demand response and energy consumption patterns for smart homes. Subsequently, the home energy management system generates optimal power schedules for individual appliances. The proposed approach is compared with similar methods for energy management to demonstrate its efficiency. The aim of article [34] is to achieve high energy efficiency through the integration of AI with Renewable Energy (RE) resources. A modified GWO algorithm is employed in the proposed energy management system for micro grids, considering modern power grid interactions. The primary objective is to regulate the power generated from renewable sources, manage their integration with the utility grid, and control battery charging and discharging based on load requirements. The proposal introduces a hybrid technique utilizing both GWO and PSO. However, the main shortcomings of the proposal include the low conversion rate and precision accuracy. Energy optimization leveraging electric vehicles has gained significant popularity recently, especially with consumers actively engaging in Peer-to-Peer energy trading, particularly involving Renewable Energy Sources (RES) and Electric Vehicles (EVs). Efficient energy management is essential for allocating energy effectively within clusters amidst complex pricing and energy exchange mechanisms. The proposal outlined in [35] utilizes deep RL for energy optimization without impacting the prosumers. It involves Peer-to-Peer energy trading with a novel hybrid mechanism and energy exchange process. Vehicle-to-Everything (V2X), including Vehicle-to-Homes and Vehicle-to-Load, is employed to minimize overall energy costs. The implementation incorporates Multi-Double Q network agents based on deep RL, alongside the utilization of Markov Decision Processes to optimize decision-making processes.

IV. CHALLENGES FOR AI

The integration of AI into Smart Grids offers unprecedented opportunities for enhancing grid efficiency, resilience, and sustainability. However, navigating the associated challenges demands concerted efforts across various factors. Challenges for AI in Smart Grids are:

- **Data Management:** Smart Grids generate vast and diverse data, requiring robust pre-processing and adaptive algorithms to ensure accuracy and relevance.
- **Security:** Cybersecurity threats, including data breaches and adversarial attacks on AI models, demand robust encryption and intrusion detection mechanisms.
- **Sustainability:** Energy-efficient AI algorithms and optimization techniques are vital for minimizing environmental impact and enhancing overall grid efficiency.
- **Interoperability:** Integration with legacy infrastructure necessitates standardization efforts and open APIs for seamless data exchange.
- **Regulatory work:** Clear regulations addressing data governance, privacy, and ethical concerns are essential for innovation while safeguarding consumer rights.

Overcoming these challenges is crucial for AI to realize its transformative potential in Smart Grids, paving the way for smarter, more resilient energy infrastructure.

V. CONCLUSION

The aim of this article is to examine recent literature concerning the intersection of SG technology and AI over the past five years. Given AI's pervasive influence across various domains, it has become indispensable in the evolution of SG. Many innovative solutions to the challenges facing SG stem from AI applications, often leveraging hybrid approaches to data management. While this article provides a concise overview of recent research, it is important to note that certain topics, such as data management, encompass a breadth of sub-issues that warrant more extensive exploration. Despite space limitations, we endeavor to offer readers a glimpse into the latest trends and technologies addressing major smart grid challenges, focusing on select papers pertinent to each issue.

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